Dense paddock-scale validation of statistical and process-based soil-moisture downscaling

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Companion validation report to "Downscaling SMIPS soil moisture to 30 m: statistical and process-based estimation of root-zone soil moisture, with validation in the Murrumbidgee catchment". This draft scaffold separates the model-development evidence from the dense-site validation evidence. Red [TODO: items] should be resolved before circulation.

Abstract

[TODO: Write after results are stabilised. Suggested structure: one sentence on the paddock-scale validation gap; one sentence describing Esdale, Tarrawarra, Nerrigundah, Llara and MRI; one sentence on the model-agnostic validation table; one sentence with headline model6/model8 skill and seasonal/wetness bias; one sentence on local calibration/spiking; one sentence on implications for deployment.]

Root-zone soil-moisture products are increasingly evaluated against sparse station networks, but management decisions occur at sub-property scale where within-paddock redistribution by terrain, soil and antecedent wetness is critical. Here we use five retained model-ready validation datasets to test whether the final statistical and process-based models developed in the companion paper preserve paddock-scale soil-moisture patterns when treated as black-box gridded predictors. The datasets span modern dense campaign points, historical high-density TDR grids and continuous probe networks. Skill is summarised using Pearson correlation (r), Nash–Sutcliffe efficiency (NSE), RMSE, ubRMSE and bias so that temporal association, error magnitude and mean-level offsets can be separated. [TODO: Insert headline values and conclusion.]

Keywords: soil moisture; dense validation; downscaling; paddock scale; model evaluation; local calibration; Nash–Sutcliffe efficiency

1 Introduction

Operational soil-moisture products generally resolve kilometres rather than management units, yet irrigation, grazing, crop establishment and ecophysiological modelling often require information at the scale of a paddock or sub-property unit. The companion paper develops two routes to 30 m root-zone soil moisture: a statistical downscaler of the ≈1 km `SMIPS TotalBucket` product using terrain, soil and meteorological features, and a process-based bucket model with a static soil–terrain–climate readout. Its central validation question is whether these models transfer across stations, years and spatial blocks in the OzNet monitoring network (Smith et al., 2012).

That station-network validation is necessary but not sufficient for the intended product. Sparse stations can test regional transfer and temporal dynamics, but they cannot directly test whether a 30 m map reproduces within-site moisture redistribution. Dense point datasets provide a "microscope" over the downscaling field: they can reveal whether residuals cluster by terrain position, wetness state or season, and whether competing models fail in the same spatial locations.

This report therefore asks three questions:

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1. **Independent paddock-scale transfer.** When evaluated at dense point observations not used for model development, how much skill do model 6 RF and model 8 process retain?
2. **Seasonal, wetness-state and terrain-conditioned bias.** Are poor predictions concentrated in particular seasons, dry/wet states or model-agnostic terrain/soil strata?
3. **Deployability through sparse local calibration.** If a landholder can provide one or a few local sensors, how much can a separate calibration layer improve local prediction while preserving independent validation?

This framing is close in spirit to two lines of previous work but differs in its validation emphasis. Process-oriented downscaling studies such as the Equilibrium Moisture from Topography, Vegetation and Soil model (EMT+VS; Ranney et al., 2015) explicitly use fine-resolution topographic, vegetation and soil information to redistribute coarse soil moisture to 10–100 m scales, and show that soil and vegetation information can improve spatial patterns when those inputs are sufficiently detailed. Recent machine-learning tools such as `mlhrsm` (Peng et al., 2024) instead emphasise application-ready high-resolution mapping, uncertainty estimation and spatial–temporal analysis from in-situ sensors, satellite products and land-surface covariates. The present report sits between these traditions: it compares a statistical and a process-based 30 m product using the same model-agnostic dense validation table, and treats seasonal bias, wet/dry compression and terrain-stratified error as primary validation targets rather than secondary diagnostics.

2 Relationship to the companion model paper

The purpose of this report is not to re-rank every model in the development ladder. Instead, it takes the two final comparison models from the companion paper and evaluates them under a different validation geometry: dense within-site observations. Table 1 summarises the division of evidence.

Table 1: How this dense validation report complements the companion model paper.

Companion model paper	**Dense validation report**
Develops the full model ladder, models 1–8.	Uses the final comparison models only: model 6 RF and model 8 process.
Evaluates transfer across OzNet stations, spatial blocks and years.	Evaluates transfer to independent dense point datasets at paddock scale.
Primary question: which validation design estimates national transfer?	Primary question: does the 30 m field preserve local spatial patterns and bias structure?
Focuses on regional/station-level skill, residual decomposition and product generation.	Focuses on seasonal/wetness bias, terrain-conditioned error, spatial clustering and sparse local adaptation.
Uses station-network evidence to justify final model choices.	Uses dense-site evidence to diagnose deployment reliability and local calibration needs.

3 Dense validation datasets

3.1 Sites and validation roles

The analysis uses five retained model-ready validation datasets, each contributing a different kind of evidence (Table 2). Esdale provides the strongest modern within-site terrain coverage but only autumn–winter dates. Tarrawarra and Nerrigundah provide unusually dense historical TDR campaign grids; because their raw points are denser than the model grid, both are evaluated after aggregation to model grid-cell support. They also carry the expected pre-`SMIPS TotalBucket` coarse-anchor caveat for model 6 RF. Llara provides the strongest multi-year seasonal validation with 32 profile-mean probes across two paddocks. MRI adds a sparser but longer continuous probe network, useful for testing whether local calibration behaviour holds when the spatial support is less dense.

Table 2: Dense validation site inventory. Values shown here reflect the current analysis cache and should be checked against the final model-agnostic tables before submission.

Site	Main role	Supports	Dates	Period	Seasons	Note
Esdale	Spatial/terrain microscope	79	9	2025-04-30–2025-07-17	Autumn, winter	Modern dense campaign; best for terrain residual inference but short seasonal coverage.
Tarrawarra	Extreme spatial-density stress test	178	19	1995-09-25–1996-11-29	Spring, summer, autumn, winter	3610 raw points aggregated to model grid cells by date; model 6 RF is affected by the historical coarse-anchor caveat.
Nerrigundah	Independent historical grid test	128	12	1997-08-27–1997-09-22	Winter, spring	Tarrawarra-like 15 cm TDR campaign aggregated to model grid cells by date.
Llara	Seasonal/temporal microscope	32	911	2022-01-01–2024-06-30	Spring, summer, autumn, winter	Profile-mean probes across WE and WW paddocks; trimmed from January 2022 for probe-establishment QC.
MRI	Sparse continuous-probe transfer	18	1799	2021-07-01–2026-06-03	Spring, summer, autumn, winter	Mulloon Rehydration Initiative profile-mean probes; trimmed from July 2021 for probe-establishment QC.

3.2 Observation harmonisation

Each dense validation dataset measures soil moisture using methods and depth supports that differ from each other and from the training datasets used for the original model development. The validation target is therefore treated as volumetric soil moisture percentage at the best available site-specific support, not as a perfectly interchangeable sensor quantity.

Llara and MRI measurements are obtained from continuously recording in-situ probes. The validation uses a profile-mean depth summary so that each timestamp contributes a single local soil-moisture estimate. Esdale measurements are handheld TDR probe readings to approximately 10 cm depth. Tarrawarra and Nerrigundah measurements are TDR campaign readings to approximately 15 cm depth.

Tarrawarra and Nerrigundah also require explicit spatial-support harmonisation. Their campaign sampling is much denser than the model prediction grid, so scoring raw probe observations would over-weight sub-pixel variability that neither a 30 m statistical model nor a 30 m process model can resolve. Validation is therefore performed after aggregating observations to the model support scale: within each model/date/grid-cell combination, observed TDR soil moisture is averaged and the prediction is sampled once from the corresponding raster cell. Residual sub-pixel

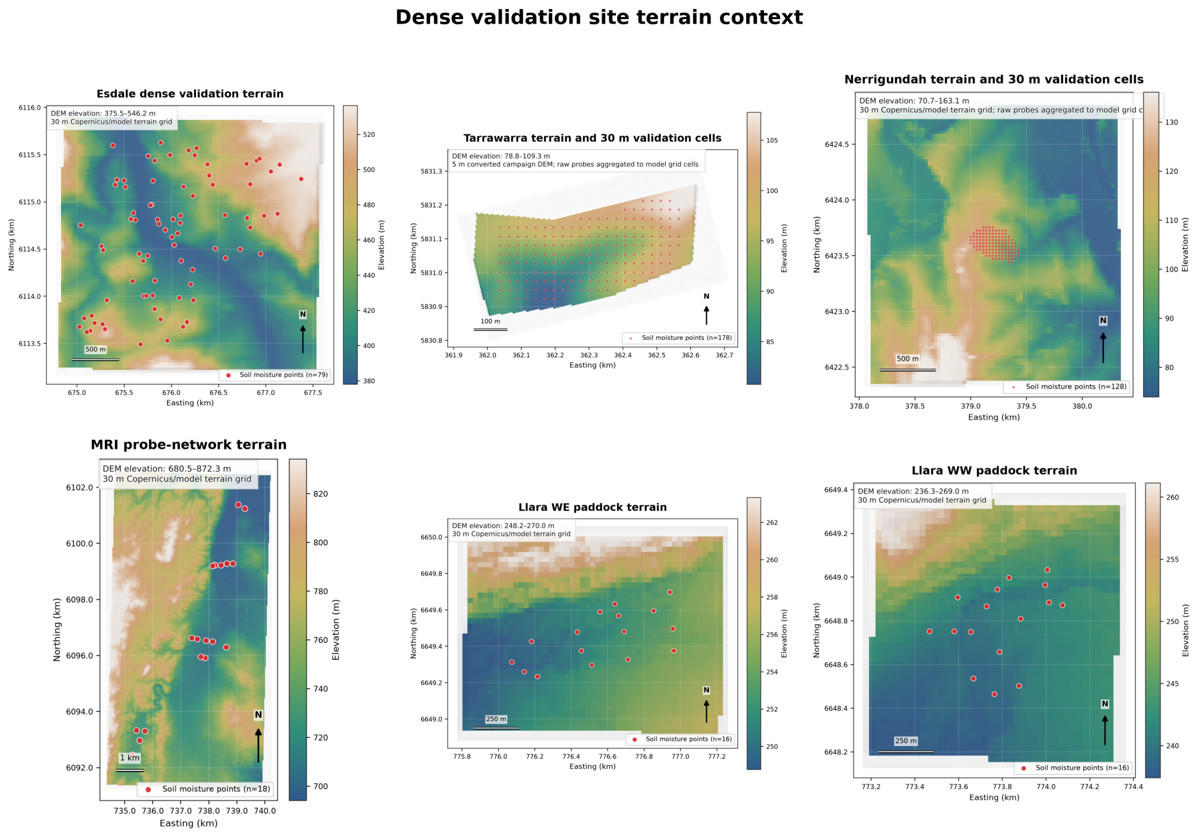

Figure 1: Available validation-site terrain context. Esdale, Nerrigundah, Llara WE, Llara WW and MRI use the 30 m Copernicus/model terrain grid; Tarrawarra uses the converted 5 m campaign DEM. Red points are soil-moisture validation supports: probe locations for Esdale, Llara and MRI, and aggregated model-grid-cell centroids for Tarrawarra and Nerrigundah.

variation is treated as an observational/support mismatch rather than direct model error. This provides an additional diagnostic: grid cells with high within-cell TDR variance mark locations where no 30 m model should be expected to score perfectly unless it explicitly represents sub-pixel structure.

3.3 Model predictions and gridded products

For each site and observation date, model predictions are generated or sampled from cached gridded products and extracted at the relevant observation support: probe coordinates for Esdale, Llara and MRI, and model grid-cell supports for Tarrawarra and Nerrigundah after aggregation. The validation protocol is deliberately model agnostic: after prediction, each model is represented only by a table of support/date predictions and associated model-agnostic covariate strata. This allows model 6 RF and model 8 process to be compared without relying on either model's internal representation.

[Caution: For Tarrawarra and Nerrigundah, the current model 6 RF prediction tables have a `SMIPS TotalBucket`-zero/coarse-anchor caveat because historical SMIPS coverage was unavailable or returned zero-valued inputs for the model runs. Do not interpret their model 6 RF skill as normal operational model6 results unless this is resolved or explicitly framed as a coarse-anchor ablation.]

4 Model-agnostic validation protocol

4.1 Common prediction table

All validation analyses begin from a common table with one row per model–support–date combination:

Table 3: Model-agnostic prediction-table schema used by all validation steps.

Field group	Required / derived columns
Identity	Site, model name, support identifier (probe or grid cell) and sampling date.
Location	Longitude, latitude and optional projected coordinates.
Target and prediction	Observed soil moisture, predicted soil moisture and residual, defined as prediction minus observation.
Temporal strata	season, year, date block, wetness state
Static model-agnostic covariates	elevation, slope, TWI, HLI, accumulation, aspect components, SLGA soil properties
Dynamic model-agnostic covariates	SMIPS summaries where available, antecedent rainfall, PET and VPD summaries

4.2 Metrics

Skill is quantified with five complementary metrics:

$$r = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2}\sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2}}, \tag{1}$$

$$\text{NSE} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \tag{2}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}, \tag{3}$$

$$\text{bias} = \frac{1}{n}\sum_i (\hat{y}_i - y_i), \tag{4}$$

$$\text{ubRMSE} = \sqrt{\max(\text{RMSE}^2 - \text{bias}^2, 0)}. \tag{5}$$

Here y_i is observed soil moisture and $\hat{y}_i$ is predicted soil moisture. We report pooled metrics by site and model, seasonal metrics, per-support metrics and paired model differences at matched support/date rows. Pearson r measures whether predictions track temporal or spatial variation, whereas the Nash–Sutcliffe efficiency (NSE; Nash and Sutcliffe, 1970) penalises both variance errors and mean-level offsets. Where observed variance is low, NSE can be unstable or harsh; interpretation therefore uses Pearson r, RMSE, ubRMSE and bias alongside NSE.

4.3 Blocking and separation of validation from calibration

Stage 1 uses all dense observations only as external validation. Stage 2 uses controlled local calibration subsets but validates on held-out points and dates. The strict spatial+temporal block is treated as the meaningful transfer estimate because it tests whether calibration learned a transferable local relationship rather than memorising a point or date.

5 Stage 1: independent dense-point validation

Before interpreting scalar validation metrics, it is useful to see examples of the gridded objects being scored. Figures 2–5 show observed dry and wet conditions where gridded products are already staged for the report. MRI is retained as a sparse continuous-probe validation in this draft, so it is shown through the DEM context map and time-series diagnostics rather than a gridded wet/dry map panel. These panels are intended as a visual orientation to the map products; all scores below are calculated from matched validation supports rather than from visual inspection.

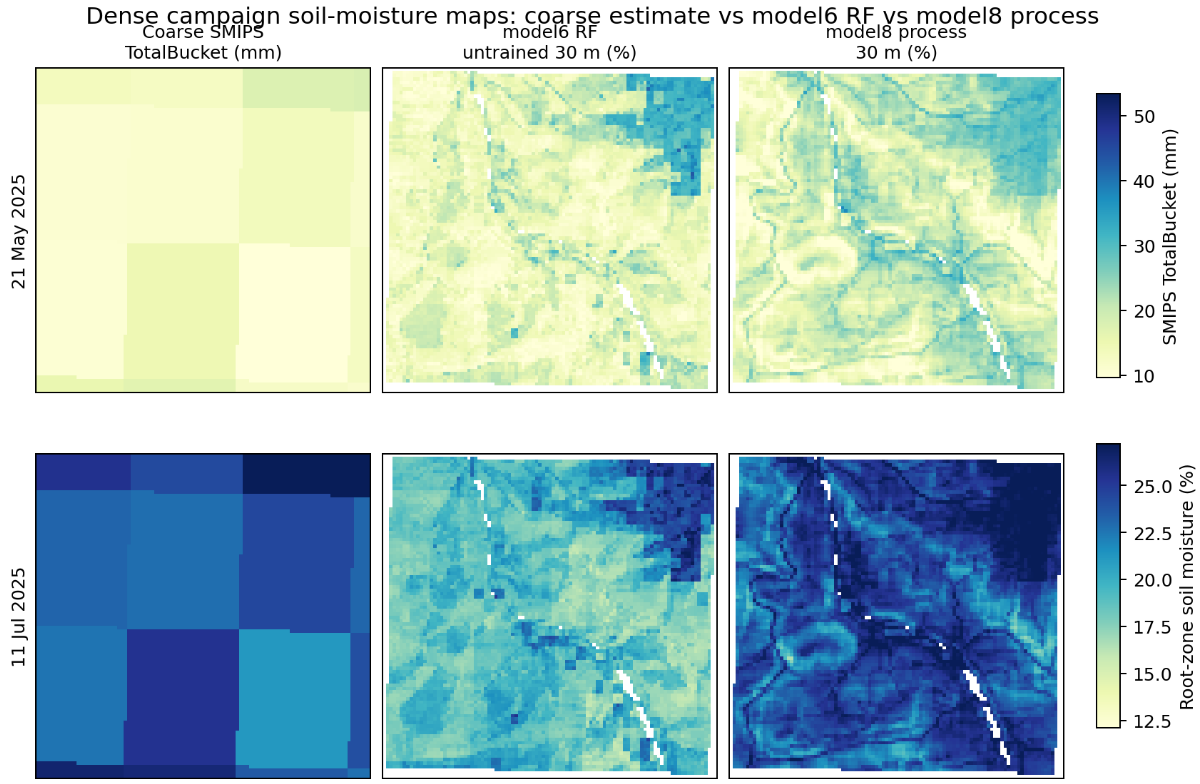

Figure 2: Esdale gridded prediction overview for the observed driest (21 May 2025; mean observed soil moisture 7.0 %) and wettest (11 July 2025; 23.2 %) dense-campaign dates. Columns show the coarse SMIPS estimate, model 6 RF and model 8 process. Mean predicted soil moisture across validation supports was 14.8 % and 15.5 % on the dry date, and 20.3 % and 24.8 % on the wet date, for model 6 RF and model 8 process respectively.

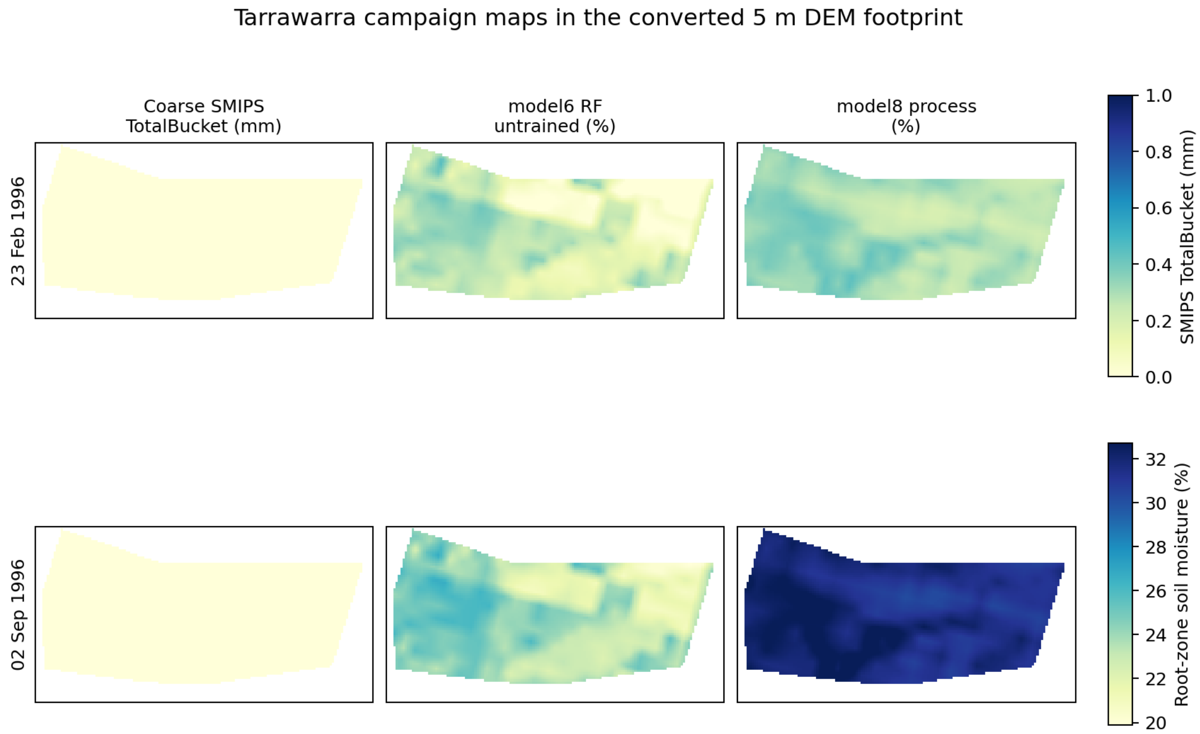

Figure 3: Tarrawarra gridded prediction overview for the observed driest (23 February 1996; mean observed soil moisture 20.3 %) and wettest (2 September 1996; 48.4 %) campaign dates after aggregation to model-grid support. Columns show the historical coarse estimate, model 6 RF and model 8 process. Mean predicted soil moisture across aggregated validation supports was 21.7 % and 23.4 % on the dry date, and 23.6 % and 31.6 % on the wet date, for model 6 RF and model 8 process respectively. [Caution: The historical `SMIPS TotalBucket` input is affected by the zero-valued coarse-anchor caveat for this run.]

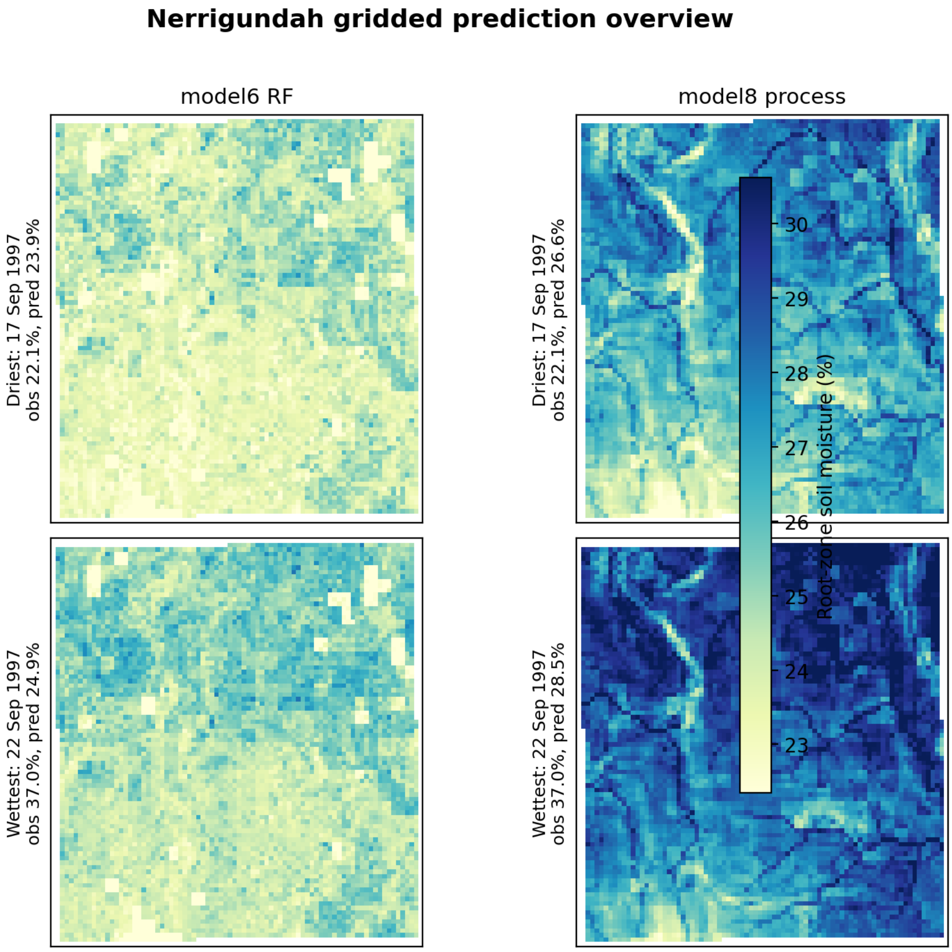

Figure 4: Nerrigundah gridded prediction overview for the observed driest (17 September 1997; mean observed soil moisture 22.1 %) and wettest (22 September 1997; 37.0 %) campaign dates after aggregation to model-grid support. Mean predicted soil moisture across aggregated validation supports was 23.9 % and 26.6 % on the dry date, and 24.9 % and 28.5 % on the wet date, for model 6 RF and model 8 process respectively.

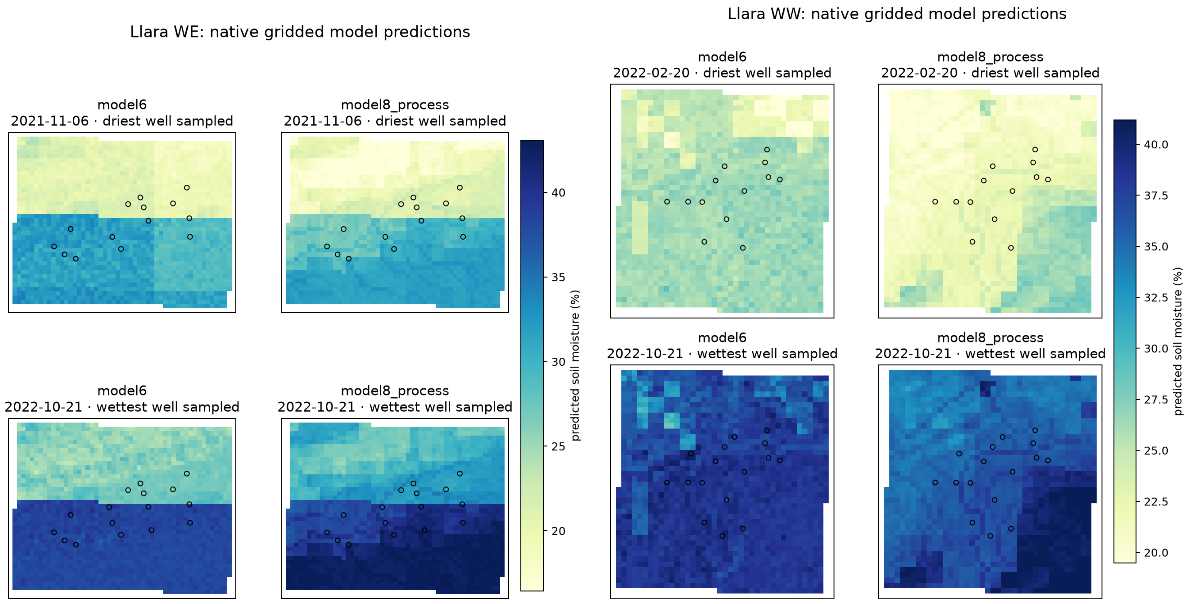

Figure 5: Llara gridded prediction examples for the WE and WW paddocks, shown separately. In WE, the dry example is 6 November 2021 (mean observed 20.9 %; model 6 RF 26.7 %; model 8 process 25.5 %) and the wet example is 21 October 2022 (67.7 %; 34.3 %; 37.2 %). In WW, the dry example is 20 February 2022 (18.4 %; 26.4 %; 21.5 %) and the wet example is 21 October 2022 (57.8 %; 38.9 %; 35.8 %).

5.1 Headline site-level skill

The first analysis pools all available support/date rows by site and model (Fig. 6). This figure should be interpreted as an external paddock-scale validation summary, not as a replacement for the OzNet cross-validation ladder in the companion paper.

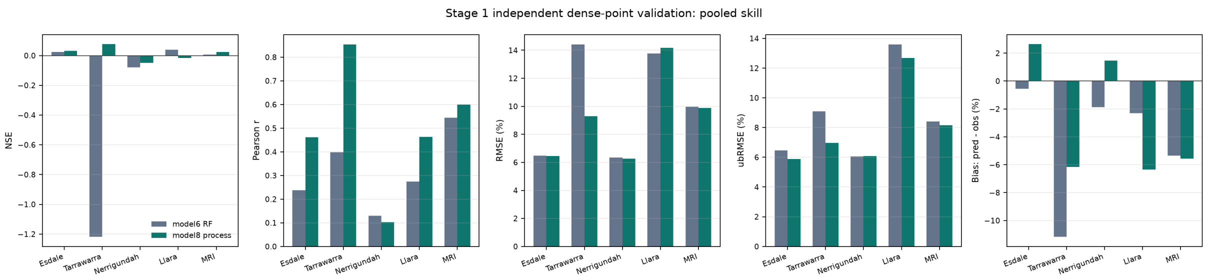

Figure 6: Stage 1 independent dense-point validation: pooled NSE, Pearson r, RMSE, ubRMSE and bias by site and model. The five metrics are reported together because correlation, efficiency, total error magnitude, bias-corrected error and mean-level offset answer different validation questions.

The headline result is that validation-site transfer remains modest at this scale, matching closely to the unseen site cross validation results from the companion paper. Across the five retained datasets, pooled NSE is near zero for most site/model combinations despite positive Pearson correlations, meaning that the models often track some spatial or temporal ordering without yet reproducing the full observed magnitude and offset. Esdale and Nerrigundah have the lowest RMSE, approximately 6–6.5 volumetric percentage points, but Nerrigundah has weak correlation for both models. Tarrawarra shows the clearest advantage for model 8 process after grid-cell support aggregation, while Llara and MRI show stronger temporal association for model 8 process but only modest differences in total RMSE. The historical campaign sites remain especially useful as stress tests of spatial structure, whereas Llara and MRI are the stronger tests of continuous temporal behaviour.

Table 4: Headline independent validation metrics from the current unified analysis cache. For Tarrawarra and Nerrigundah, n is the number of model/date/grid-cell supports after aggregation, not the number of raw TDR observations. Llara and MRI are QC-trimmed to 2022-01-01 and 2021-07-01, respectively.

Site	Model	n	NSE	r	RMSE	ubRMSE	Bias / interpretation
Esdale	model 6 RF	560	0.023	0.238	6.477	6.453	−0.558; weak spatial skill, small mean dry bias.
Esdale	model 8 process	560	0.031	0.462	6.452	5.884	2.645; higher association but wet-biased.
Tarrawarra	model 6 RF	2154	-1.219	0.398	14.402	9.093	−11.169; strong dry bias, interpreted with the `SMIPS TotalBucket`-zero caveat.
Tarrawarra	model 8 process	2154	0.077	0.853	9.291	6.958	−6.157; strongest Tarrawarra association after support aggregation.
Nerrigundah	model 6 RF	1536	-0.079	0.130	6.341	6.054	−1.887; low RMSE but weak association under the historical-grid test.
Nerrigundah	model 8 process	1536	-0.050	0.102	6.254	6.078	1.476; slightly lower RMSE but still weak association.
Llara	model 6 RF	28390	0.039	0.274	13.777	13.582	−2.308; slightly better error/bias than model 8 process overall.
Llara	model 8 process	28390	-0.017	0.463	14.172	12.677	−6.335; stronger association but larger dry bias.
MRI	model 6 RF	29046	0.006	0.545	9.972	8.413	−5.354; moderate temporal association with dry bias.
MRI	model 8 process	29046	0.024	0.600	9.882	8.157	−5.577; slightly better efficiency/correlation, similar dry bias.

5.2 Predicted and observed time series

Spatially averaged time series provide a diagnostic of seasonal tracking and mean-level bias. They are not a substitute for support-level scoring, but they make temporal drift and wet/dry compression visually apparent.

5.3 Seasonal and wetness-state bias

The dense validation should emphasise seasonal and wetness-state bias because a model can have acceptable pooled error while systematically overpredicting dry states or underpredicting wet

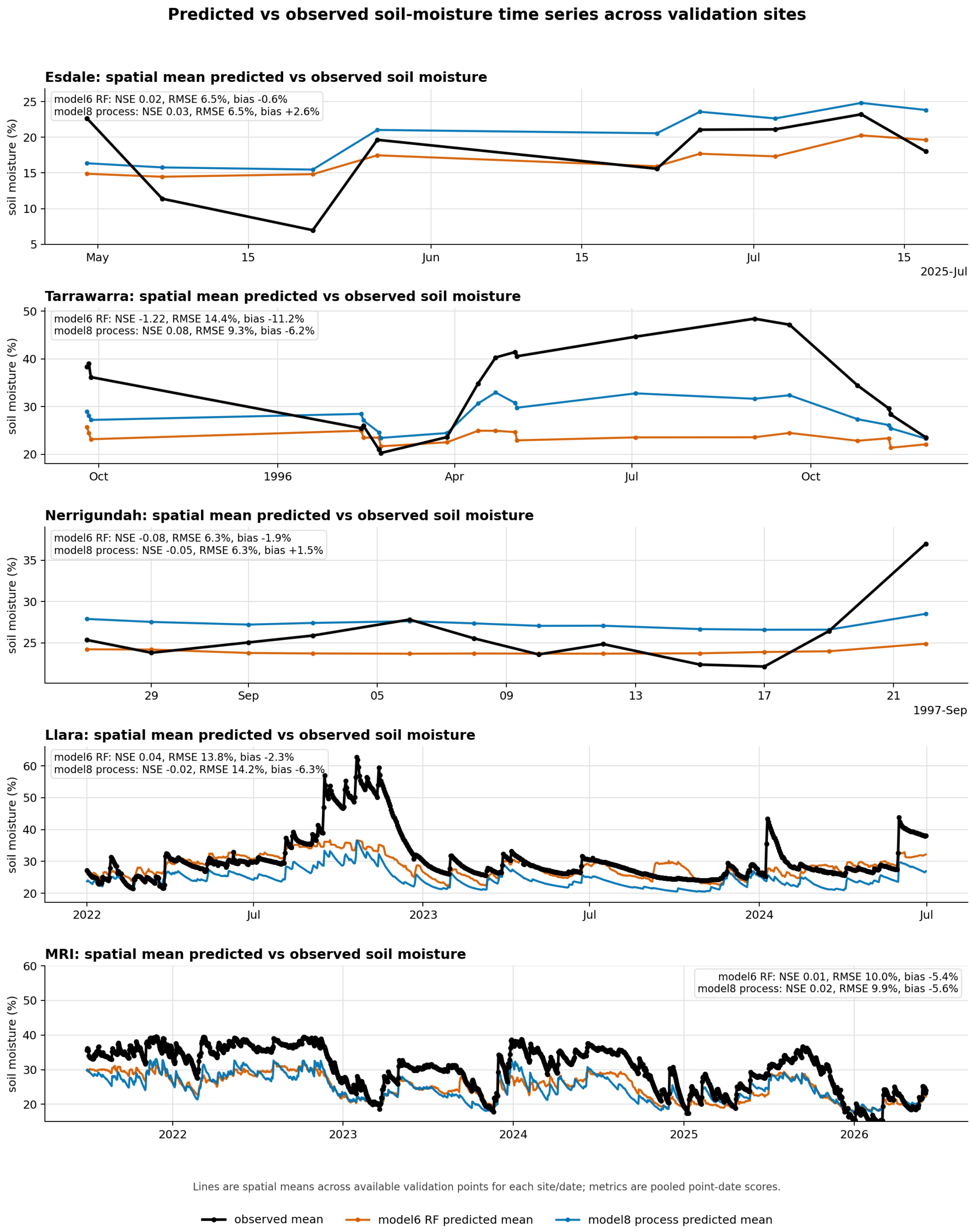

Figure 7: Observed and predicted soil-moisture time series by site. Values are spatial means over available validation points for each date. The MRI panel is scaled to 15–60 % soil moisture to keep the observed and predicted traces readable across the long probe record.

states. This is particularly relevant to trafficability, drought stress and paddock management decisions.

Seasonal residuals show that mean bias is not stable through time (Fig. 8). Esdale is limited to autumn–winter, where model 6 RF changes from a small autumn wet bias to a winter dry bias, while model 8 process remains wetter than observed in both seasons. Tarrawarra is dominated by the model 6 RF historical coarse-anchor caveat; model 6 RF is strongly dry in spring, autumn and winter but close to neutral in summer, while model 8 process is still dry in spring, autumn and winter and wet-biased in summer. Nerrigundah provides only winter–spring campaign dates, so it is best read as an independent historical grid check rather than a full seasonal diagnostic. Llara and MRI provide the strongest multi-year seasonal tests; both show persistent dry bias in the continuous-probe records, especially for model 8 process.

Wetness-state residuals indicate a general compression of the observed moisture range (Fig. 9). The models tend to overpredict the driest observed quartile and underpredict the wettest observed quartile, with the effect particularly visible in the longer probe records and the historical campaign grids. This pattern is important because a map that gets the mid-range approximately right may still misrepresent the very wet and very dry paddock positions that often drive management decisions.

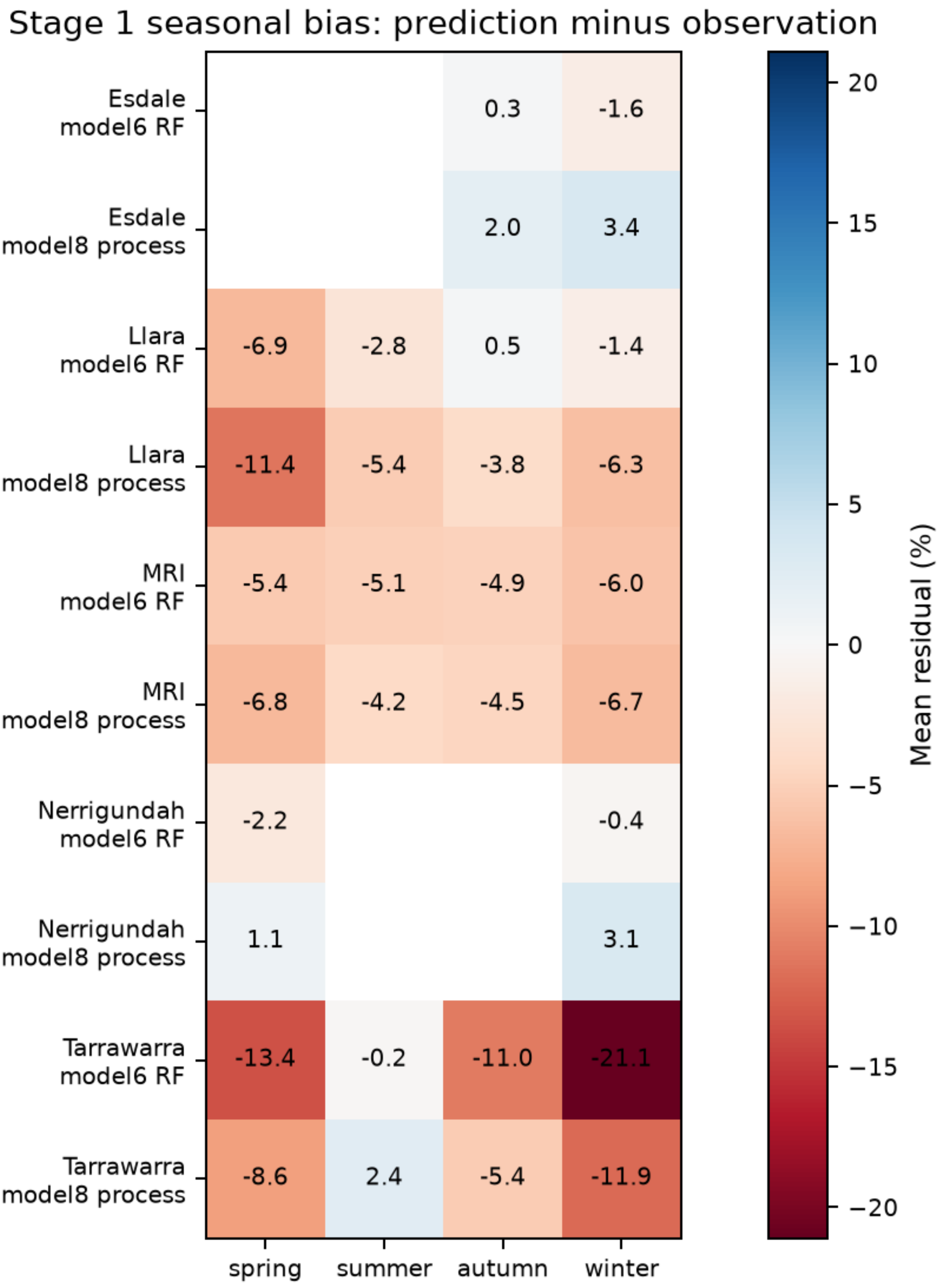

Figure 8: Seasonal bias by site and model. Residuals are prediction minus observation; positive values indicate overprediction. Esdale and Nerrigundah have limited seasonal coverage, while Llara and MRI provide the clearest multi-year seasonal bias diagnostics.

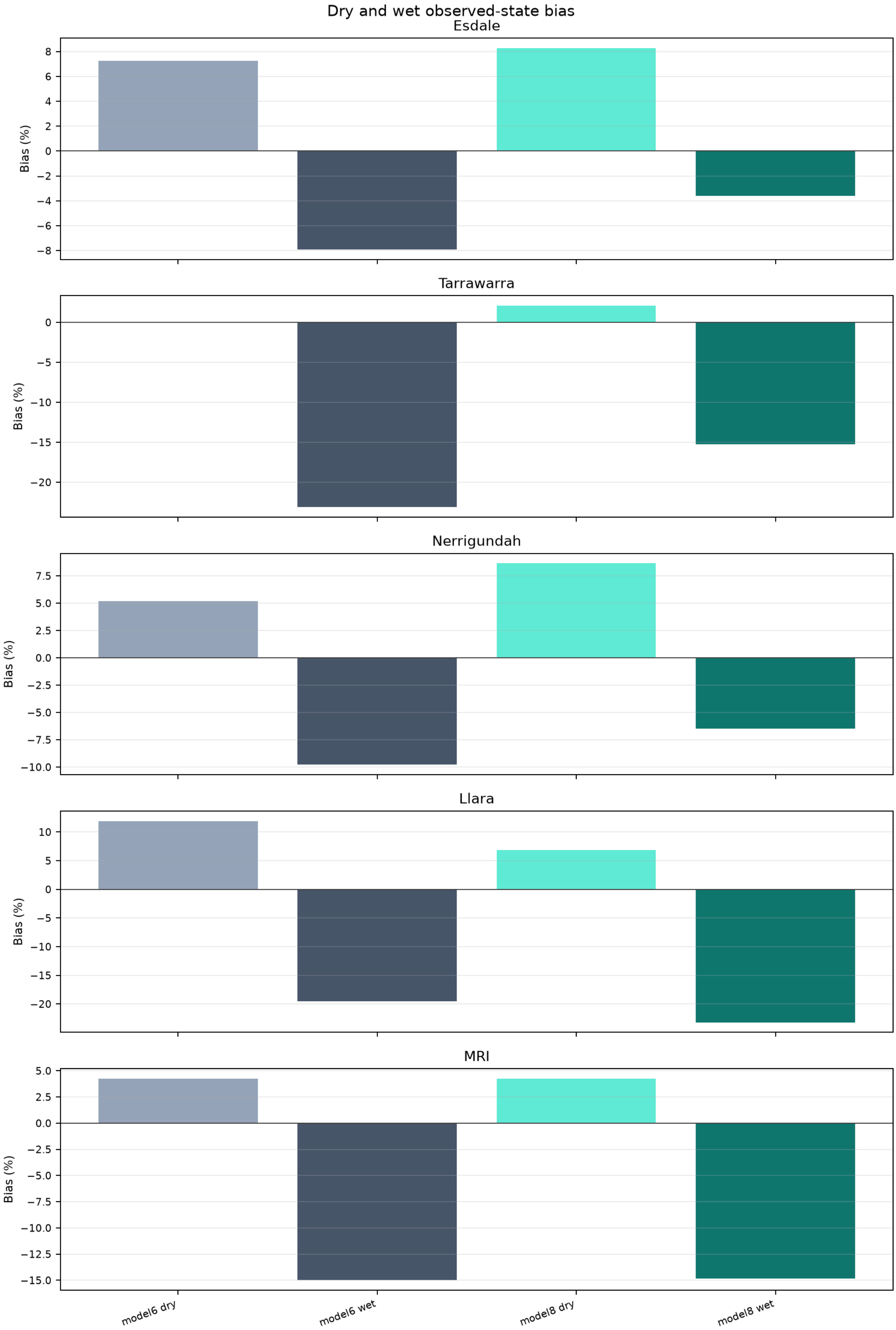

Figure 9: Dry/wet observed-state bias. The driest and wettest observed quartiles test whether models compress extremes. The main diagnostic is whether predictions move toward the centre of the observed distribution, i.e. overpredicting dry supports and underpredicting wet supports.

5.4 Terrain and model-agnostic covariate stratification of error

The validation supports allow error to be stratified by model-agnostic covariates that drive the gridded predictions, including elevation, slope, TWI, HLI, accumulation, aspect components, SLGA soil properties and antecedent meteorological summaries. The main text should show only the most notable patterns; complete strata tables belong in the supplement.

The global-scaled static terrain–soil PCA asks whether the validation sites occupy different absolute terrain–soil settings (Fig. 10). This view preserves site means, so between-site separation should be interpreted as environmental transfer distance rather than within-site model failure. PC1 explains 32.4 % of the pooled static terrain–soil variation and mainly contrasts sandier, higher-elevation, higher-bulk-density supports against clayier, high-HLI and east-facing supports. PC2 explains 20.2 % and is dominated by soil available water capacity, eastness, HLI, bulk density, slope and TWI. PC3 explains a further 14.6 % and loads most strongly on TWI and accumulation, which is a reminder that the two-axis PCA figure is a useful orientation, not a complete terrain-space representation.

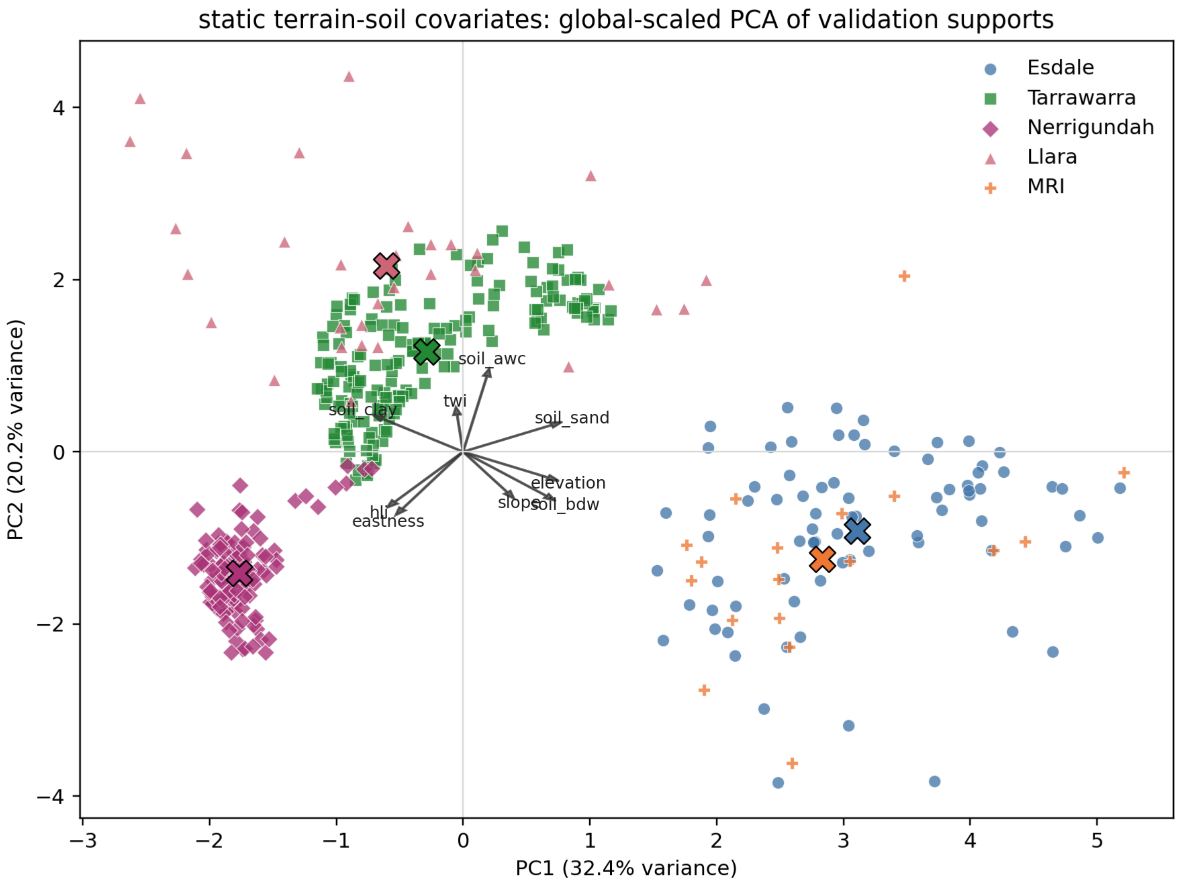

Figure 10: Global-scaled static terrain–soil PCA of validation supports. PCA inputs are elevation, slope, aspect components, TWI, HLI, accumulation and SLGA soil properties. Colours identify sites and arrows show the dominant loadings, so the figure describes how the retained validation sites occupy different terrain–soil covariate space before any residual overlay is added.

The site-standardised PCA removes each site's mean terrain–soil offset before fitting the PCA (Fig. 11). This is the cleaner view for asking whether high-error supports occupy analogous within-site positions, for example local ridges, hollows, high-HLI positions or soil-texture extremes. This view should be interpreted as a diagnostic map of terrain failure modes, not as a causal attribution model; the PCA axes are rotations of correlated covariates.

After removing site-level covariate means, residuals show only weak-to-moderate alignment with common terrain/covariate gradients. The strongest direct association in the current diagnostic tables is HLI versus support-level bias at approximately $r = 0.375$, which corresponds to roughly 14 % of variance in a simple univariate sense. The static terrain–soil PCA does not show a clean “bad supports here, good supports there” separation, and PC1+PC2 explain only about 41 % of the site-standardised static terrain–soil variation. Several stronger associations appear on PC3 or in model/site-specific combinations, and the covariate ranking is diffuse across exposure/aspect, slope, soil texture, TWI and accumulation. This suggests that model error is partly structured, but that the structure is site-dependent rather than a universal terrain-error mechanism shared across all validation sites. Given the introduced historical coarse-anchor caveat for model 6 RF at Tarrawarra and Nerrigundah, the two models also show no large, simple difference in terrain-space clustering of error.

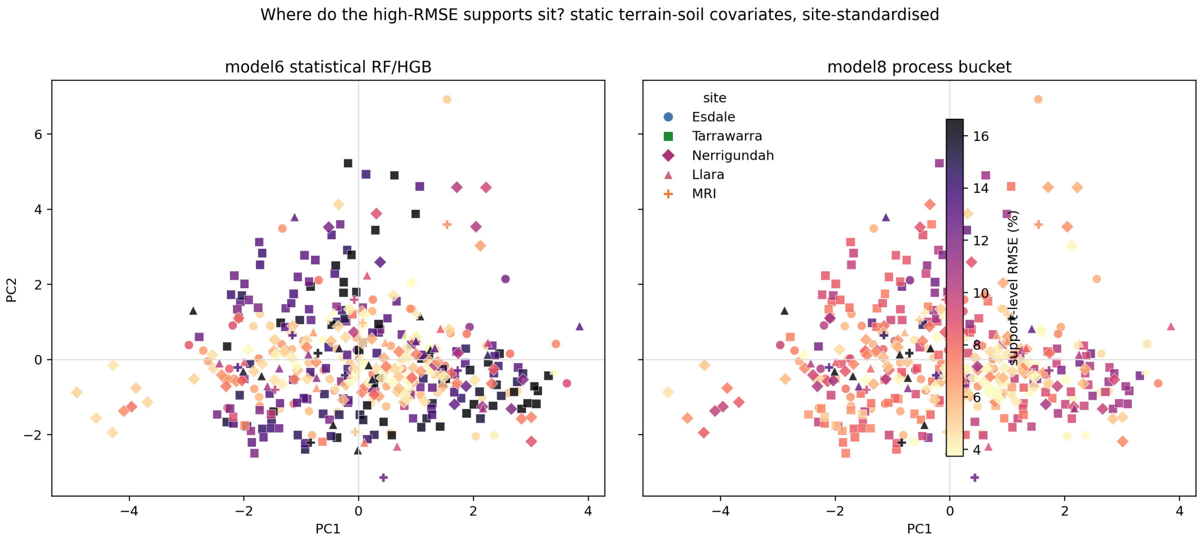

Figure 11: Site-standardised static terrain–soil PCA with supports coloured by support-level RMSE. Within each site, static covariates are z-scored before PCA, so local terrain contrasts are comparable across sites even when the absolute site environments differ.

5.5 Spatial clustering of prediction quality

Prediction-quality maps test whether poor performance is spatially organised. For publication, show support-level maps of NSE, Pearson r, RMSE, ubRMSE and bias where sample size permits, plus a paired model-advantage map where useful. Interpolated quality surfaces can aid visual diagnosis but should be labelled as diagnostic interpolation, not model output.

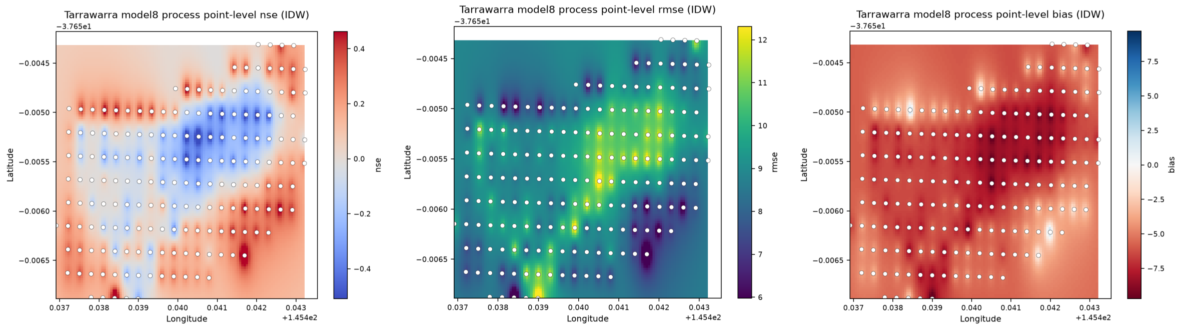

Figure 12: Example spatial prediction-quality surfaces for Tarrawarra model 8 process. These are diagnostic interpolations of validation metrics at aggregated model-grid-cell support, not gridded model predictions. [TODO: Replace or extend with a final multi-site figure that uses consistent colour scales and includes paired model-advantage maps.]

6 Stage 2: sparse local calibration and local spiking

6.1 Rationale

Independent validation tells us where the national/global model fails. A deployment question follows: can a small amount of local information correct site-level or terrain-conditioned bias without requiring dense sampling at every property? This stage should be interpreted as a local adaptation experiment, not as independent validation of the base models.

The experiment should mimic realistic local information. A new landholder will usually not know where the model performs badly, but may know where the landscape is generally wet, dry, low, high, clayey or exposed. Calibration point selection should therefore compare random selection against simple landscape-knowledge proxies and model-informed extremes.

6.2 Calibration layers and point-selection strategies

The current calibration layers are:

1. **Bias offset:** one constant residual correction.
2. **Seasonal offset:** separate residual offsets by season where enough dates exist.
3. **Affine correction:** local linear rescaling of prediction level.
4. **Residual ridge layer:** residual model using selected model-agnostic covariate covariates.

Point budgets are three, five, ten, 25 %, 50 % and all eligible local calibration points. Validation is performed on held-out points and dates, with the strict spatial+temporal block treated as the primary transfer estimate.

[TODO: State exactly which selection strategies are retained in the final protocol. Current set: random, landscape wet/dry prior, global-prediction extremes, and the field-knowledge wet/dry proxy upper bound.]

6.3 Learning curves

The learning curves summarise whether sparse local calibration improves strictly held-out supports as more local information is supplied. To match the minimum-budget analysis in Table 5, the main figures show the best deployable non-random, non-proxy strategy at each point budget: either the landscape wet/dry prior or global-prediction extremes, excluding the observed wet/dry proxy. RMSE gain is useful because it has the same units as soil moisture (Fig. 13), but it can hide whether the calibrated model becomes efficient relative to the observed variance. The corresponding NSE learning curve (Fig. 14) therefore shows whether calibration moves the model toward modest but interpretable efficiency thresholds such as NSE ≈ 0.2–0.3. Random placement is retained as an appendix comparator because it was generally ineffective under the strict block.

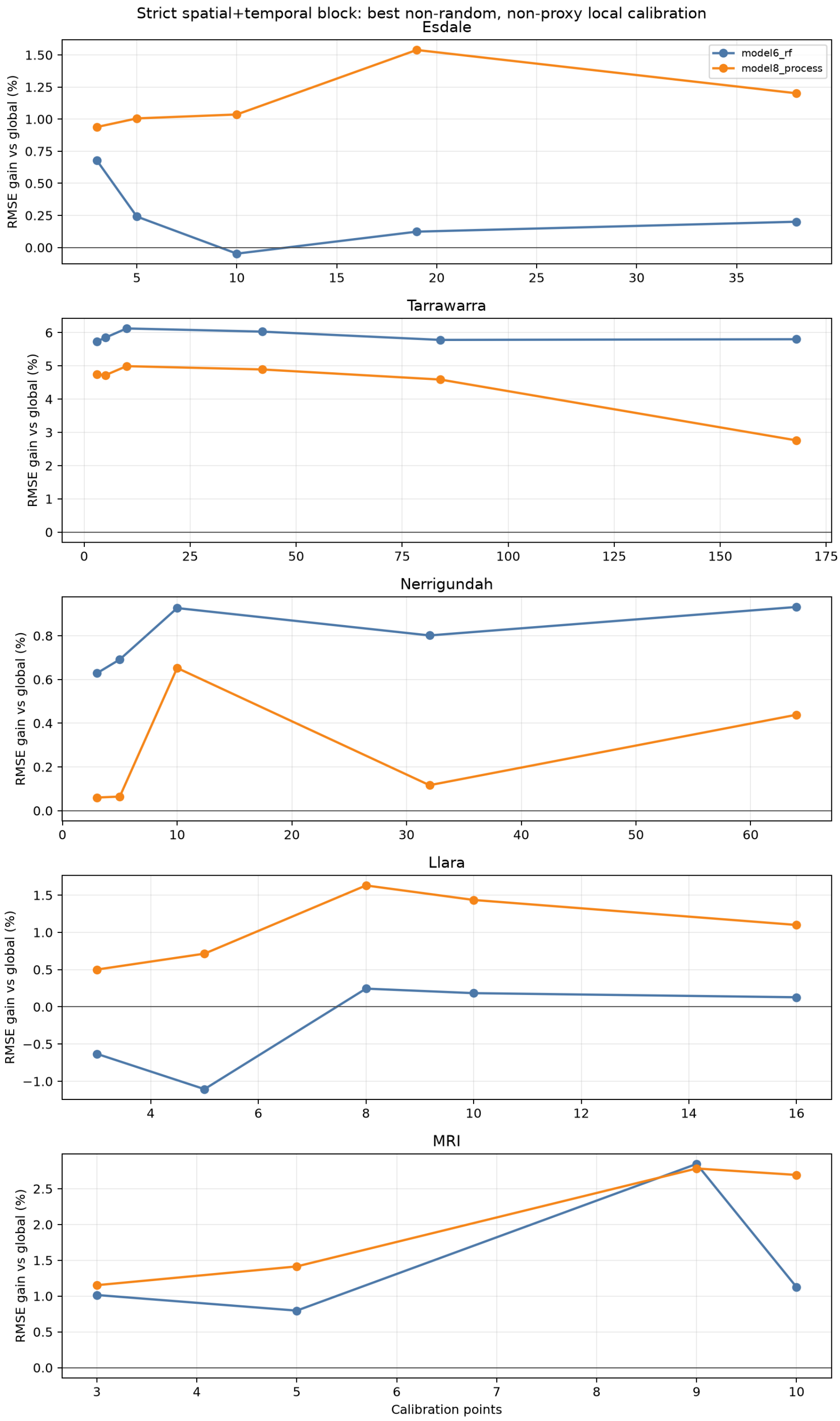

Figure 13: Prior-guided sparse local-calibration learning curves under strict spatial+temporal blocking. Each point is the best deployable non-random, non-proxy strategy/method combination for that site, model and local point budget. Positive values indicate lower RMSE after calibration.

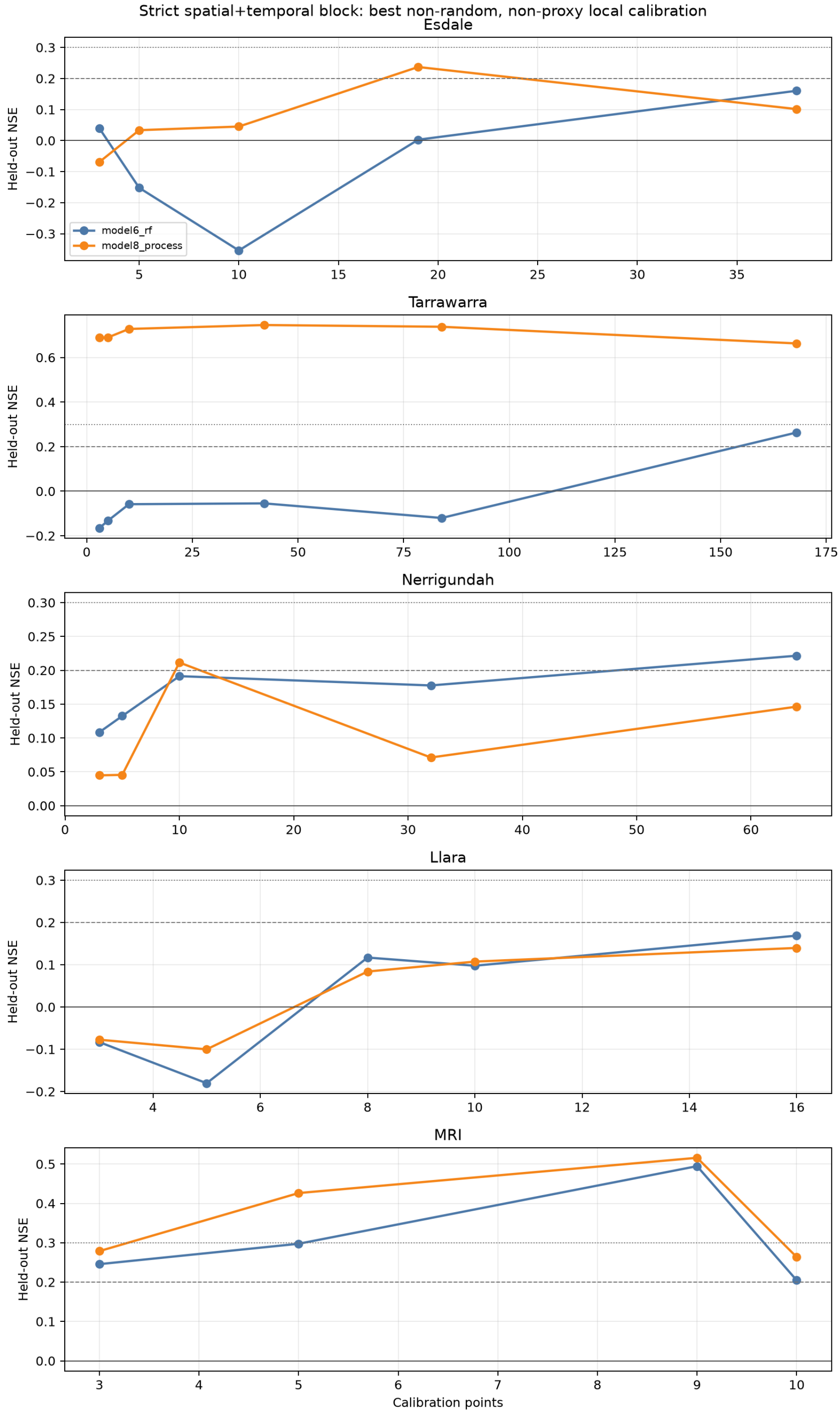

Figure 14: Prior-guided sparse local-calibration learning curves expressed as held-out NSE under strict spatial+temporal blocking. Each point is selected using the same deployable non-random, non-proxy strategy set as Table 5. Dashed and dotted reference lines mark NSE = 0.2 and 0.3, respectively. These thresholds are not universal performance targets, but they are useful for asking how much local information is needed before calibration becomes meaningfully better than the uncalibrated baseline.

6.4 Minimum local information budgets

The deployment question is framed as a minimum information budget: how many local supports are needed before strict held-out NSE reaches approximately 0.2 or 0.3? Table 5 uses the current strict spatial+temporal block and excludes the `field_knowledge_wetdry_proxy` upper-bound strategy because that proxy uses observed chronic wet/dry behaviour. The result is mixed rather than uniformly optimistic. Tarrawarra model 8 process responds strongly with three landscape-prior supports, MRI reaches NSE = 0.3 with five to nine supports, and Nerrigundah reaches NSE = 0.2 but not 0.3. Esdale model 6 RF and both Llara models remain below 0.2 under the deployable prior-guided designs in this cache. The practical sensor-placement story should therefore emphasise parsimonious but defensible local information budgets, not the assumption that any small sensor set will fix within-property failure modes.

Table 5: Minimum local calibration supports needed to reach modest held-out NSE thresholds under the strict spatial+temporal block. Entries report the smallest non-proxy strategy/method combination reaching the threshold, or the best current result if the threshold is not reached.

Site	Model	NSE ≥ 0.2	NSE ≥ 0.3
Esdale	model 6 RF	Not reached; best 38 supports, global-prediction extremes + residual ridge (NSE = 0.160).	Not reached; same best design (NSE = 0.160).
Esdale	model 8 process	19 supports, landscape wet/dry prior + residual ridge (NSE = 0.237).	Not reached; best NSE = 0.237.
Tarrawarra	model 6 RF	168 supports, global-prediction extremes + residual ridge (NSE = 0.263).	Not reached; best NSE = 0.263.
Tarrawarra	model 8 process	3 supports, landscape wet/dry prior + affine correction (NSE = 0.689).	3 supports, landscape wet/dry prior + affine correction (NSE = 0.689).
Nerrigundah	model 6 RF	64 supports, landscape wet/dry prior + residual ridge (NSE = 0.222).	Not reached; best NSE = 0.222.
Nerrigundah	model 8 process	10 supports, landscape wet/dry prior + residual ridge (NSE = 0.211).	Not reached; best NSE = 0.211.
Llara	model 6 RF	Not reached; best 16 supports, global-prediction extremes + bias offset (NSE = 0.169).	Not reached; same best design (NSE = 0.169).
Llara	model 8 process	Not reached; best 16 supports, global-prediction extremes + bias offset (NSE = 0.140).	Not reached; same best design (NSE = 0.140).
MRI	model 6 RF	3 supports, global-prediction extremes + bias offset (NSE = 0.246).	9 supports, global-prediction extremes + residual ridge (NSE = 0.495).
MRI	model 8 process	3 supports, global-prediction extremes + bias offset (NSE = 0.279).	5 supports, global-prediction extremes + residual ridge (NSE = 0.426).

6.5 Do process and statistical models respond differently?

The process-vs-statistical comparison is both technical and philosophical. A statistical downscaler may already encode nonlinear terrain and soil relationships learned from the station network, but it may be brittle outside the training distribution. A process model may have simpler temporal dynamics and fewer degrees of freedom, but a local layer may correct its site-level state more cleanly. The validation datasets allow this difference to be measured as the marginal gain from the same sparse local information.

In the current strict prior-guided experiment, process-vs-statistical responsiveness is site dependent rather than a single model ranking (Fig. 15). model 8 process often gains more than model 6 RF at Esdale and Llara, while Nerrigundah tends to favour larger gains for model 6 RF and MRI is mixed. Tarrawarra remains a special case: model 6 RF begins with a large dry bias linked to the historical coarse-anchor problem, so simple local calibration can remove part of that offset even when model 8 process remains the lower-error model after calibration. A clearer mechanistic interpretation will require comparing the calibration supports against model-agnostic diagnostics such as TWI, HLI, elevation, soil texture, antecedent rainfall/PET and the availability or failure mode of SMIPS/process state variables.

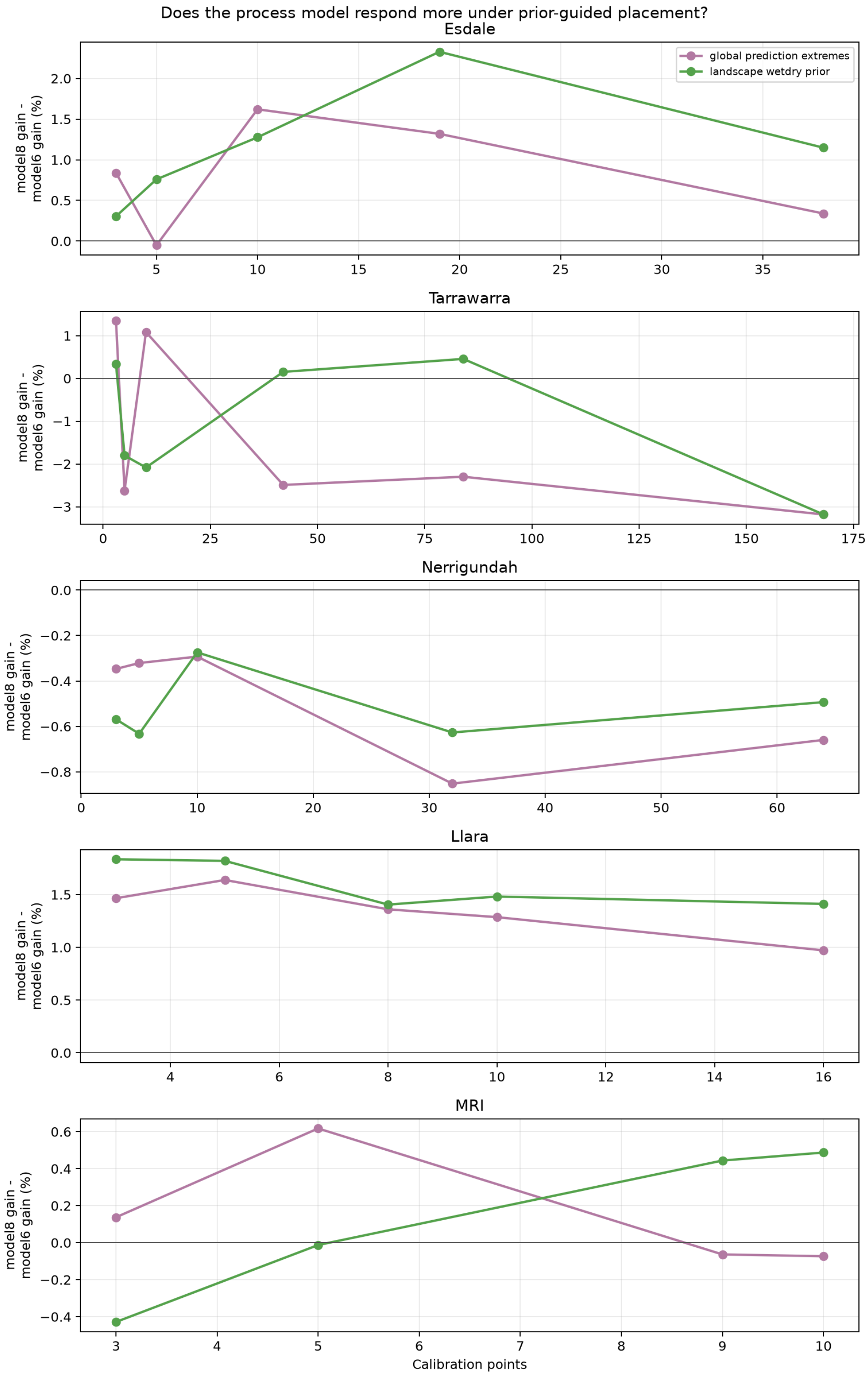

Figure 15: Process-vs-statistical local calibration responsiveness under deployable prior-guided placement. Positive values indicate that model 8 process gained more RMSE improvement than model 6 RF from the same non-random placement strategy and sparse local information budget under strict spatial+temporal blocking.

7 Discussion

7.1 What dense validation adds beyond station validation

[TODO: Synthesis paragraph. Suggested claim: the companion paper establishes regional/station transfer; dense validation reveals whether the 30 m product has meaningful within-site spatial structure and whether residuals organise by terrain, season and wetness state.]

7.2 Model complementarity at paddock scale

[TODO: Discuss whether model 6 RF and model 8 process fail in the same places. Use paired comparisons and maps. Avoid a single global winner unless the paired evidence is strong and caveats are resolved.]

7.3 Implications for local deployment

[TODO: Discuss the sparse-sensor deployment story. Frame local calibration as a separate adaptation layer placed on top of a national/global model, not as leakage into independent validation.]

7.4 Future public training and validation data

[TODO: Discuss likely value of additional public dense datasets, sensor networks, campaign data and remote/proximal sensing products. Separate data that improve model training from data that improve independent validation.]

8 Limitations

- Esdale has strong spatial/terrain coverage but short autumn–winter coverage.
- Tarrawarra and Nerrigundah have exceptional raw point density; validation now aggregates those points to model grid cells by date before scoring. Their current model 6 RF `SMIPS TotalBucket`-zero/coarse-anchor caveat prevents direct interpretation as operational model 6 RF performance. High within-cell TDR variance should be treated as support mismatch unless future models explicitly represent sub-pixel structure.
- Llara has strong temporal/seasonal coverage but only 32 probes across two paddocks.
- MRI has a long continuous record but only 18 retained probe supports, so its local-calibration results are sensitive to sparse-network geometry.
- Soil-moisture depth definitions and profile integration differ across datasets and must be documented carefully.
- The validation datasets remain limited in climate and soil coverage relative to the intended national product.
- NSE can be unstable when observed variance is small; Pearson r, RMSE, ubRMSE and bias must be interpreted alongside it.
- Local calibration experiments are deployment sensitivity analyses, not independent validation of the base models.

9 Conclusions

[TODO: Write as 4–6 numbered conclusions. Suggested structure: (1) dense validation is a distinct evidence layer; (2) headline independent skill and caveats; (3) seasonal/wetness/terrain bias; (4) model6/model8 complementarity; (5) sparse local calibration implications; (6) priority next data.]

Code and data availability

All validation code and report-generation scripts are maintained in the `DMM_validation` repository. Model definitions and the companion paper are maintained in the `DownscalingMoistureModel` repository. [TODO: Add repository URLs, branch names, commit hashes, data access statements and any restrictions before submission.]

Supplementary material plan

- Full model-agnostic prediction tables for Esdale, Tarrawarra, Nerrigundah, Llara and MRI.
- Full per-site/model metric tables.
- Full terrain/model-agnostic covariate strata tables.
- Full paired support/date model comparison tables.
- Local calibration design tables, selected calibration points and replicate-level learning-curve outputs.
- Additional prediction-quality maps and gridded dry/wet map examples.

A Random-placement local calibration sensitivity

Random point selection is retained as a conservative deployment comparator. It is useful precisely because it is weak: if random placement fails while prior-guided placement improves skill, the result supports the argument that local calibration requires some landscape knowledge rather than arbitrary sensor placement. Under the strict spatial+temporal block, random placement generally improves RMSE more reliably than NSE, and Esdale remains especially difficult (Figs. A1–A3).

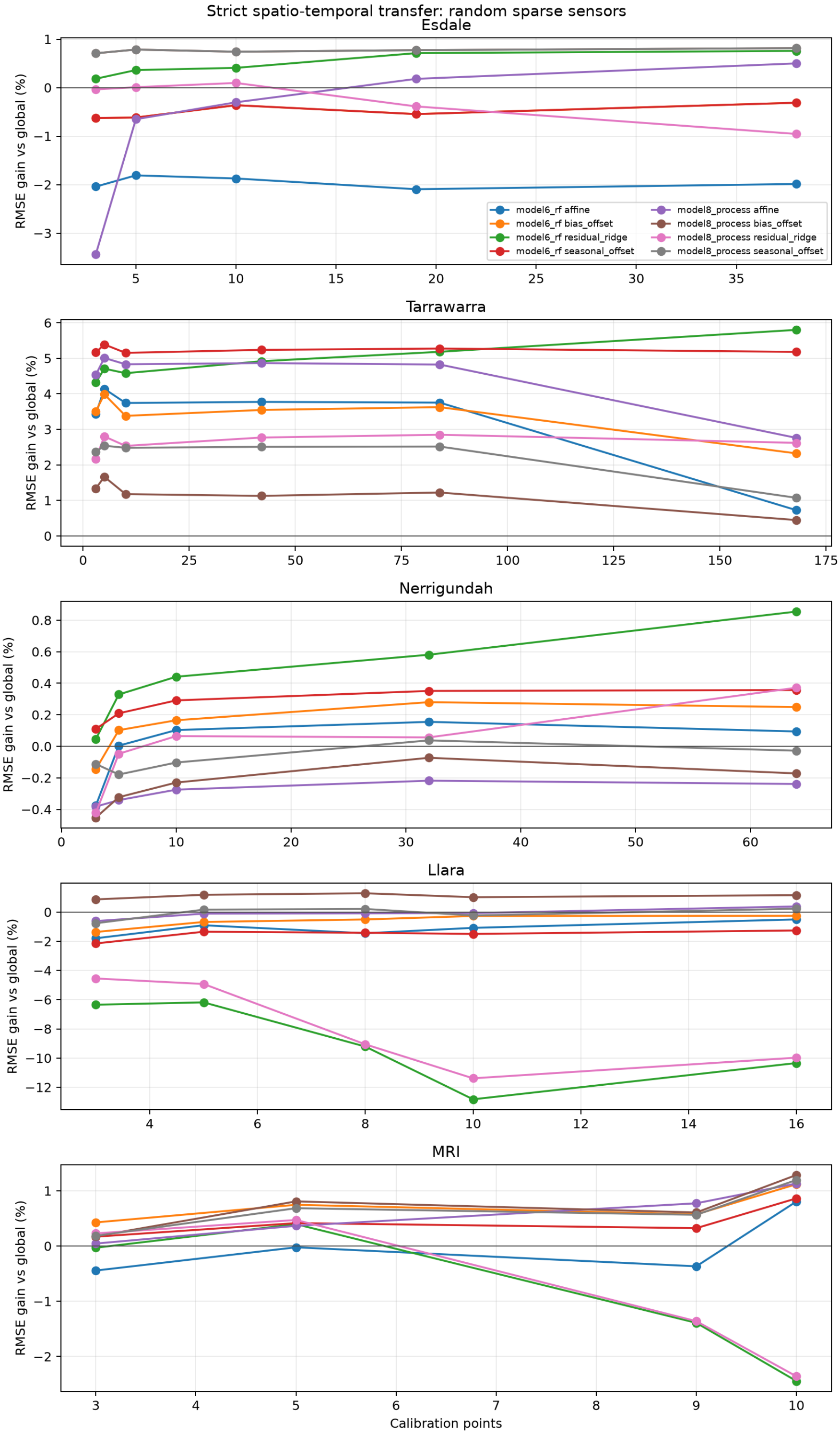

Figure A1: Appendix random-placement sparse local-calibration learning curves under strict spatial+temporal blocking. Positive values indicate lower RMSE after calibration.

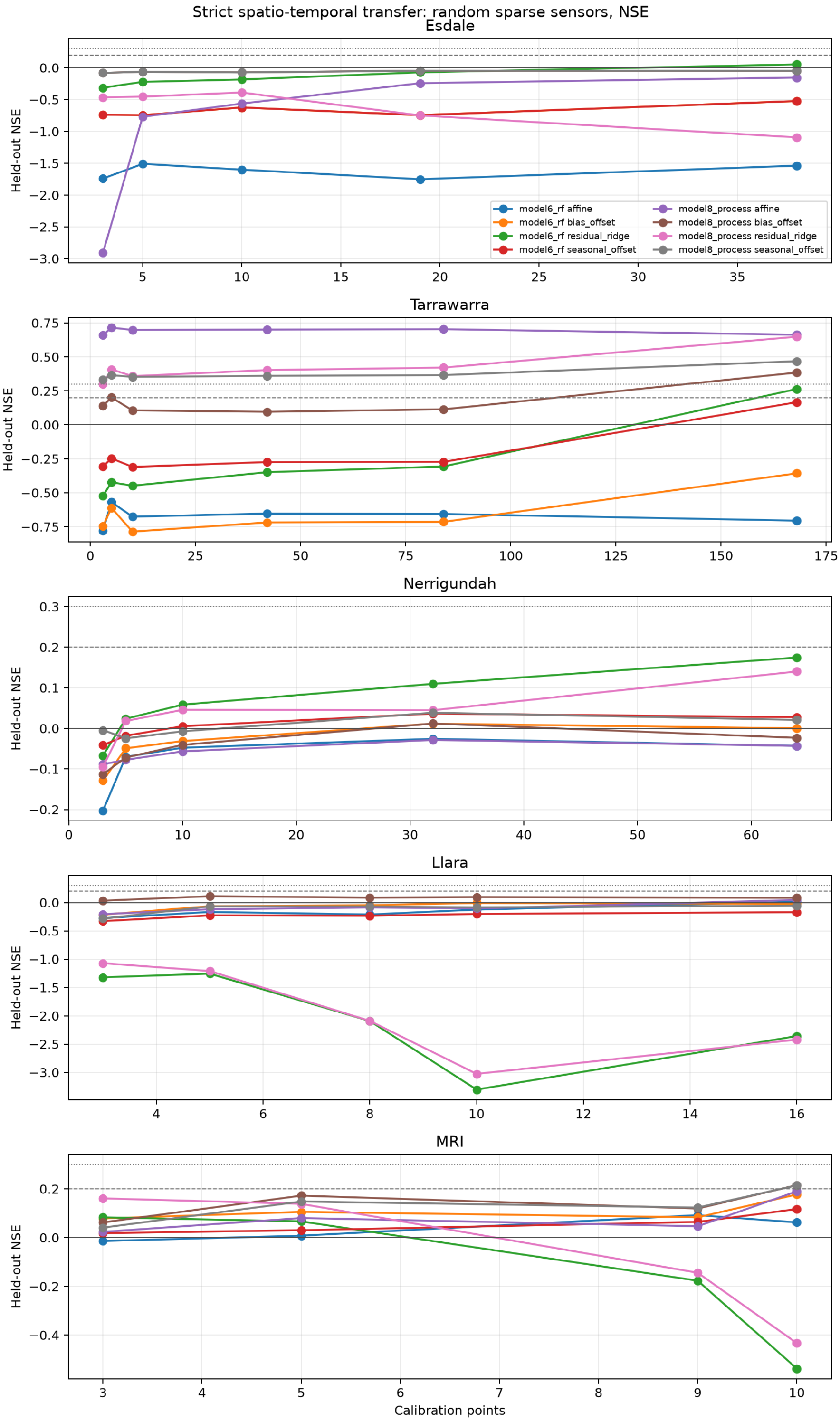

Figure A2: Appendix random-placement sparse local-calibration learning curves expressed as held-out NSE under strict spatial+temporal blocking.

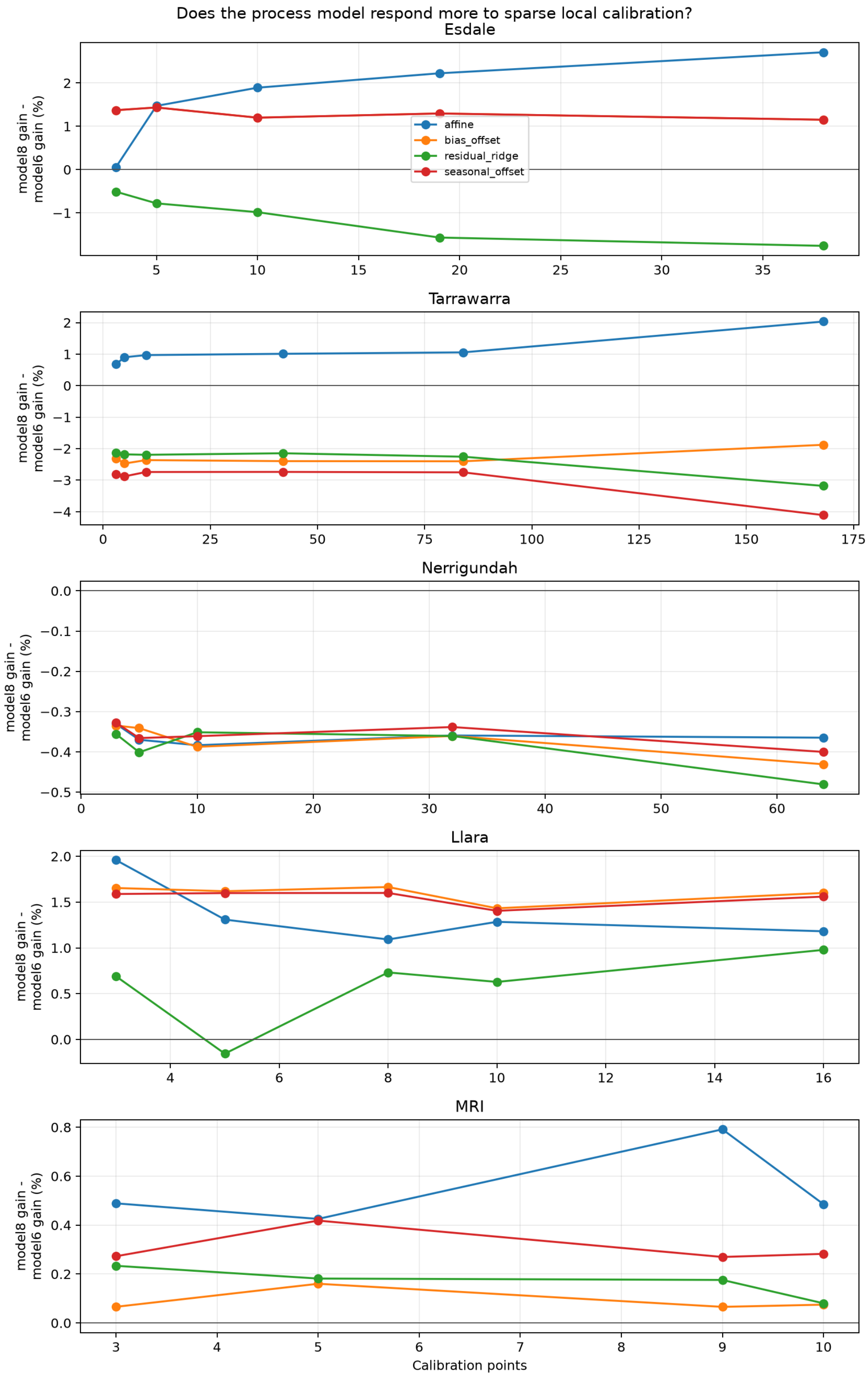

Figure A3: Appendix process-vs-statistical local calibration responsiveness under random placement. Positive values indicate that model 8 process gained more RMSE improvement than model 6 RF from the same random sparse local information budget.

B Dynamic model-agnostic covariate PCA

The main text uses static terrain–soil PCA to keep the site-space argument focused. The corresponding global-scaled PCA for dynamic model-agnostic covariates is retained here because it folds in terrain, soil, coarse SMIPS context and antecedent meteorological summaries. It therefore helps separate static terrain–soil separation from broader terrain+climate transfer distance, without overloading the main terrain-residual figure.

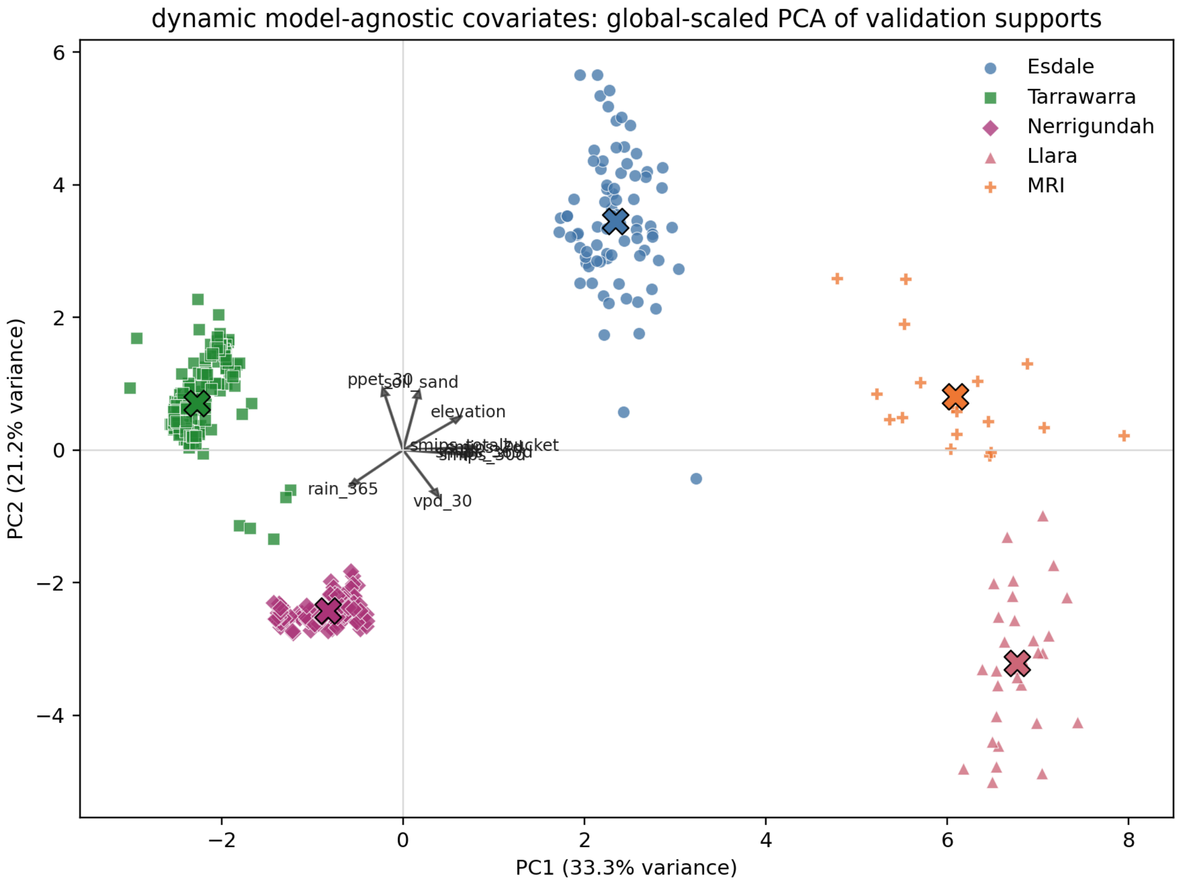

Figure A4: Appendix global-scaled PCA of dynamic model-agnostic covariates. The feature set combines static terrain–soil variables with coarse SMIPS summaries, antecedent rainfall, PET and VPD summaries. Colours identify validation sites and arrows show the dominant loadings.

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